Integrated HP 21xx I/O platform: design review

Revision **0.9 — 2026-11-08 — accepted BOM choices and rail indicators**.

This document defines the current design of the integrated HP 21xx I/O board. The first application is HP 13210/7900 disk emulation; the platform supports other peripheral functions through a compatible STM32 application/FPGA image selected on SD. Review the design before schematic capture.

An initial four-sheet KiCad capture and PDF preview are now available. Capture notes and remaining circuit work distinguish the wired first pass from the unreleased portions of this design.

Component choices, pin assignments and configuration structure below are the current design. Remaining decisions and verification work are collected in **section 5**. Electrical planning allowances still need validation for the completed circuit and each supported application.

The power calculation, functional BOM and its quantity/sourcing notes, STM32 pin inventory and FPGA pin inventory accompany this document. The BOM includes selected devices, proposed support parts and explicit unresolved items. It is a design/procurement baseline, not a schematic-derived assembly release. Through-hole components are sourced from **TME**; surface-mount parts use **JLCPCB/LCSC** where matched. **HRO TYPE-C-31-M-12/C165948** is supplied and soldered by JLCPCB in SMT assembly, as explicitly selected. All installed components must remain within the **12.8 mm** envelope.

The HP bus workbook and STM32 workbook provide signal-allocation references. The prototype KiCad board provides the board outline and card-edge geometry for layout.

1. Overview / goals / problem statement

The goal is a **generic HP 21xx I/O bus module that can implement different types of peripheral interface** on a common hardware platform. Selectable FPGA gateware supplies the bus behavior; an optional STM32 application supplies storage, control and user interfaces. The wire-wrap area allows application-specific external circuitry without redesigning the board. Support for any particular interface still depends on the available pins, electrical/timing limits and the corresponding gateware/software.

The user shall be able to **dynamically select an application and its matching FPGA gateware from the pendant**, using images stored on SD, and run the selected peripheral function on the board.

The present wire-wrapped carrier-board prototype proves the initial approach but is not a reproducible integrated platform. The new board should consolidate the electronics, provide dependable startup/recovery and external development access, and select compatible images from SD without rewriting unchanged code. HP 13210/7900 disk emulation is the first application, not the hardware's scope.

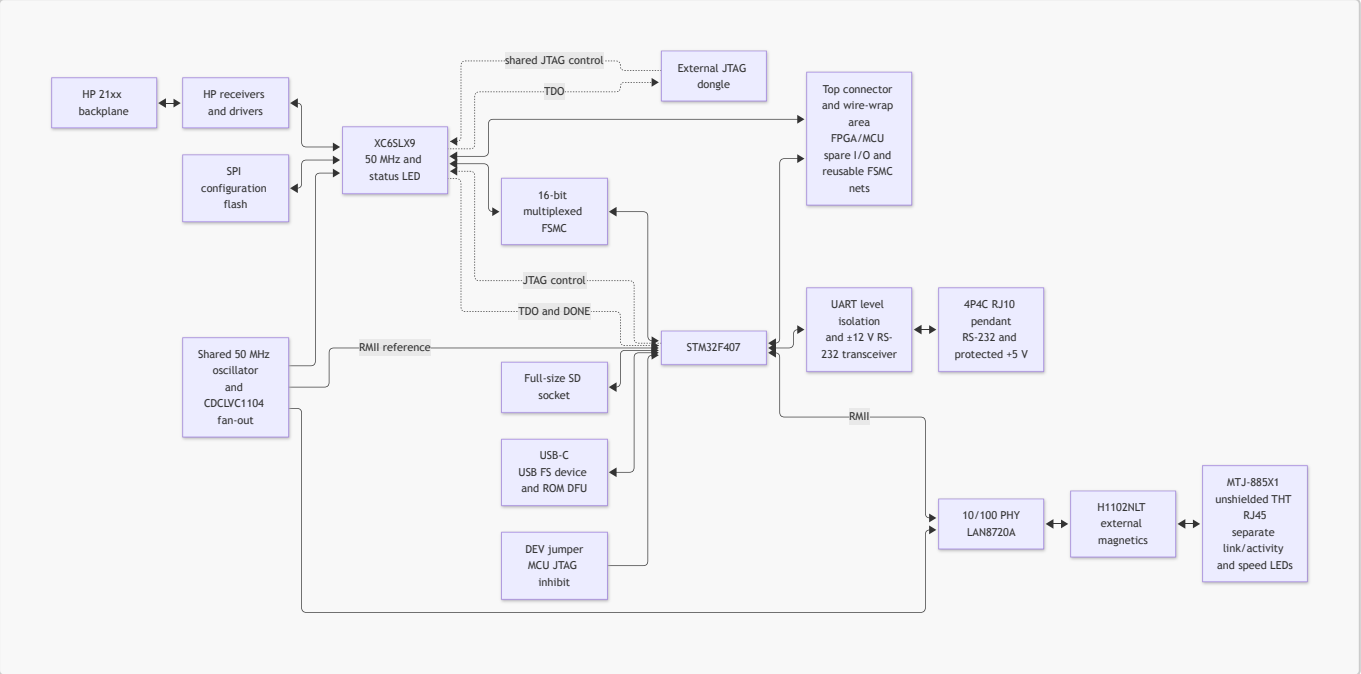
1.1 Current decisions

Item	Current design direction
Construction	One purpose-designed PCB replacing the wire-wrapped carrier-board assembly.
FPGA	**XC6SLX9-2TQG144C, LCSC C27408**, ISE target `xc6slx9-2-tqg144`.

Item	Current design direction
FPGA clock and indication	Shared 50 MHz source, buffered to P56. **P143 active high: FPGA pin → LED/resistor series path → GND**.
Processor	**STM32F407VGT6, LCSC C12345**, LQFP100/**1 MiB** flash.
Storage	Full-size SD, four-bit SDIO; **PC7 card present**, **PB1 write protect**.
USB	**HRO TYPE-C-31-M-12/C165948**, supplied and soldered by JLCPCB, with **USBLC6-2SC6, C21713975** protection, for USB communications and factory-ROM DFU. VBUS detection required, away from PA9.
Pendant	**4P4C/RJ10**, exactly four conductors: +5 V, GND, RS-232 TX and RX. MCU **PA9/PA10 USART1** also supports debugging.
HP electrical interface	Reuse the receiver/driver circuits from HP21xxReceiverDriver, incorporating its corrected 3.3 V driver-input bias.
PON	Resistor/diode conditioning to FPGA PROGRAM_B, with power-aware gating and one BAT54 isolating the MCU reset button.
FPGA management	STM32 PC0/PE2/PC2/PC3 connect to FPGA TCK/TMS/TDI/TDO; PE3 reads DONE. A local 14-pin FPGA JTAG connector supports an external dongle.
DEV mode	A jumper indicates external FPGA development: the STM32 loader must perform **no FPGA JTAG activity at all**. Applications must honor the same ownership state.
Boot platform	Pendant supports dynamic application/gateware selection from SD. Resident STM32 SD loader validates the selected pair, avoids rewriting matching images, and supports checking/updating the FPGA image against the selected profile.
Recovery/debug	BOOT0 selection, STM32 reset button and **SWD-only MCU debug** on PA13/PA14. Disable MCU JTAG and SWO/parallel trace; PA15/PB3/PB4 are expansion GPIO. Factory ROM DFU installs/recovers the resident SD loader.
Expansion	Top connector/wire-wrap exposes spare FPGA and STM32 GPIO, PA4/PA5 general expansion pins, and FSMC nets for reuse after full MCU release. FPGA P66/P62 remain included.
Ethernet	STM32 **RMII** → LAN8720A → **Pulse H1102NLT/C18584 external magnetics** → **Adam Tech MTJ-885X1 unshielded THT RJ45**, sourced from TME; separate board LEDs; 10/100 Mb/s, no Gigabit requirement. TME orderability remains open.
Mechanical height	**5/8 inch (15.875 mm ≈ 15.9 mm) card pitch**, **1.6 mm PCB**; allow **at most 12.8 mm above the component-side PCB surface**, accounting for the adjacent card's solder-side leads.
Power	HP 4.85 V / 50 A and −2 V / 15 A machine supplies reported available. Separate MCU regulator fed through Schottkys from HP and USB; ±12 V available for RS-232; +28 V unused.

Item	Current design direction
Schematic organization	Four **A3** sheets: FPGA; STM32 and its peripherals; HP connector/receivers/drivers; top connector/expansion/wire-wrap.

2. High-level architecture



The FPGA implements bus timing and peripheral-specific logic; STM32 firmware implements higher-level behavior and SD storage. Retain the existing FSMC transport for compatibility. A selected application and FPGA design must agree on their register protocol before the application starts servicing the HP bus. An application that does not use STM32 transport may instead leave the complete MCU FSMC connection in high impedance and use those FPGA pins for expansion.

The prototype host is an HP 2100S with 32K words and disk select codes **14/15 octal**. Those are the disk application's present configuration, not a restriction on future platform applications. HP-model-dependent signals such as RUN, BIOS/BIOI/BIOO and SFSB require explicit profile support.

Four-sheet A3 KiCad organization

Sheet	Contents and circuit ownership
1. FPGA	XC6SLX9-2TQG144C, shared 50 MHz oscillator/fan-out, SPI configuration flash, active-high P143 LED, mode/HSWAPEN/SUSPEND straps, INIT_B/DONE access, FPGA JTAG connector and FPGA-side supply regulation/decoupling. FPGA ends of HP/FSMC/expansion nets are here.
2. STM32	STM32F407VGT6, independent 8 MHz crystal, USB-C/protection, full-size SD with PC7 CD and PB1 WP, RS-232/4P4C, Ethernet PHY/RJ45-magnetics and reference-clock receiver, MCU LED, SWD/BOOT0/DEV/reset, Schottky-OR MCU regulator and power sensing. MCU JTAG/FSMC ends and reset isolation/gating are here.
3. HP interface	HP edge connector, all 43 receivers and 24 drivers, their bias/termination networks, HP rail entry and the raw-PON conditioning network.

Sheet	Contents and circuit ownership
4. Expansion	Top connector, uncommitted FPGA pins including P66/P62, uncommitted STM32 GPIO including PA4/PA5, spare RS-232 channels, reusable FSMC nets, optional link/driver provisions, and small wire-wrap area with labelled supply/ground access.

Keep SD, USB, RS-232 and their support on the STM32 sheet, as requested; do not split them into separate functional sheets. Power/reset circuitry is owned by its relevant sheet rather than creating another power page. A simple way to keep **four drawing sheets total** is to make the FPGA drawing the root and the other three child sheets. **All four schematic sheets shall use A3 size**; arrange the circuits within those sheets during schematic capture. Use explicit hierarchical interfaces for shared buses and reset signals, and consistent supply names. A signal exposed on the expansion drawing is the same net as at the FPGA, not a second pin allocation. Physical wire-wrap area and connector placement must be reserved in the PCB outline too.

3. Hardware

The following sections group the reviewed hardware by interface and device. Open implementation details and verification work are listed in section 5.

3.1 Interfaces

Each row describes an **external entity and its connection to the module**. The MCU-to-FPGA JTAG, FSMC and DONE links belong in the device sections.

External entity	Module connection	Purpose / detailed section
HP 21xx computer	86-contact HP card edge; bus receivers/drivers and supply entry	Peripheral interface to the computer; 3.2.5 and Appendix A
Ethernet LAN	RJ45 10/100BASE-TX, transformer isolated; no PoE	Network access for supported applications; 3.3.8
USB host computer	USB-C, USB 2.0 full-speed device and VBUS input	Communications and MCU recovery power/DFU; 3.3.1/3.3.3
Removable full-size SD card	SD socket with mechanical present/write-protect switches	Platform profiles, images and application data; 3.3.4
External control/debug pendant	Four-conductor 4P4C/RJ10: +5 V, GND, RS-232 TX/RX	UI/buttons and serial debugging; 3.3.5
Experimental circuitry / external peripheral	Top connector and wire-wrap posts for available FPGA/MCU signals, labelled supply/ground	Application-specific extension; 3.2.4/3.2.6/3.3.9
External service/programming tools	Local 14-pin 2.54 mm FPGA connector and right-angle 20-pin 2.54 mm ST-LINK/SWD header	External maintenance access; 3.2.3.2 and 3.4

HP machine capacity is reported as **4.85 V / 50 A** and **−2 V / 15 A**. The board may also use ±12 V for RS-232 and future analog conditioning; +28 V is unused. Per-slot contact/copper current and pendant allowance remain to be checked. USB host operation/USB PD and Ethernet PoE are not requested.

3.2 FPGA

Selected **Xilinx XC6SLX9-2TQG144C, LCSC C27408**, commercial grade, TQG144, speed −2; use ISE target `xc6slx9-2-tqg144` and the pin inventory below. Preserve **P143 active-high LED**: FPGA pin → LED/resistor series path → GND. The accepted LED is **KT-0603R/C2286**, with **470 Ω/C23179**; verify current and brightness; the disk gateware currently uses activity pulses and heartbeat.

The configuration guide, electrical data sheet and installed `xc6slx9_tqg144.bsd` package file are the pin/supply references.

3.2.1 Power / regulators

Supply / load	Voltage	Current requirement / allowance	Regulation
FPGA VCCINT	**1.2 V** nominal	**1,000 mA provisional allocation**, actual design/startup demand to be estimated	Independent **2 A buck** from HP 4.85 V
FPGA VCCAUX and VCCO	**3.3 V** selected	**200 mA provisional FPGA allocation**, excluding the external pull-ups below	Shared `+3V3_HP` rail, independent **2 A buck** from HP 4.85 V
Receiver collector pull-ups	3.3 V	**645 mA** maximum resistor calculation: `43 × 3.3 / 220 Ω`	Included in `+3V3_HP`
Driver-input pull-ups	3.3 V	**39.6 mA**: `24 × 3.3 / 2 kΩ`	Included in `+3V3_HP`
Flash, LED, PON, other bias/reserve	3.3 V	Approximately **79 mA** allocation	Included in `+3V3_HP`
Shared 50 MHz oscillator and fan-out	3.3 V	**50 mA** planning allowance	Included in `+3V3_HP`
FPGA/interface subtotal before Ethernet	3.3 V	**1,013 mA** planning load	Shared `+3V3_HP`
Ethernet PHY, magnetics drive and LEDs	3.3 V	**200 mA planning allowance**, including 10 Mb/s fallback case	Same HP-side rail
Total shared HP-side 3.3 V rail	**3.3 V**	**1,213 mA ≈ 1.21 A**	**2 A buck**

Use **TI TLV62569DBVR, C141836, SOT-23-5** for both stages: 2.5–5.5 V input, adjustable output, 2 A rating. Set one to 1.2 V and one to 3.3 V. These are engineering current allocations, not measured FPGA consumption or guaranteed worst-case chip currents. Use Xilinx Power Estimator for the actual gateware,

clocks, temperature and I/O loads, then verify startup and operation. Connect every VCCINT/VCCAUX/VCCO pin and provide local bypassing. FPGA supply requirements, regulator data sheet.

The pull-ups make a material load: a 3.3 V LDO would dissipate about **1.88 W** at the revised 1.213 A shared rail load; a buck at an assumed 90% efficiency loses about **0.44 W**. The 1.2 V/1 A stage would lose 3.65 W as an LDO from 4.85 V. Use switching regulators; check their inductors, effective capacitance, copper and thermal behavior. The five-pin DBVR has no power-good output.

Detailed resistor currents, feedback values, inductor calculations, −2 V termination load and direct +5 V driver consumption are in the power calculation. Estimated board 4.85 V input subtotal is approximately **2.13 A including Ethernet**, before HP output collector currents and pendant load. It is not a complete worst-case bound. Power/reset sequencing must account for the flash readiness delay and independent MCU power domain.

Use the same selected support circuit for all three TLV62569 stages: **Bourns SRP5030T-2R2M / C1329375**, 2.2 µH; **4.7 µF X7R input**; **100 nF input bypass**; **22 µF X5R output**; 1% feedback resistors and an EN-to-VIN 0 Ω link. The detailed part numbers, per-rail dividers, quantities, inductor-current checks and layout rules are in power section 3.1.

3.2.2 Clock generator

Use **one free-running 50 MHz source**, **YXC OT2EL4C4JI-111OLP-50M, C5203552**, 3.3 V CMOS, SMD3225, feeding a **TI CDCLVC1104PWR, C485518** 1:4 clock fan-out buffer on +3V3_HP. Fit 100 nF locally at each IC and a 4.7 µF buffer supply bypass. Pull oscillator OE high with 10 kΩ; tie the buffer's 1G enable high. No MCU firmware or FPGA configuration is needed to start this clock.

CDCLVC1104 connection	Destination
CLKIN pin 1	Oscillator output pin 3
Y0 pin 3	FPGA P56 application clock
Y1 pin 8	LAN8720A XTAL1/CLKIN
Y2 pin 5	STM32 PA1 RMII REF_CLK
Y3 pin 7	Leave unconnected
1G pin 2 / VDD pin 6 / GND pin 4	High / +3V3_HP / GND

Use separate point-to-point clock branches, with source-series footprints (start with 10 Ω at buffer outputs; finalize with routing/load analysis). Match PHY/MAC clock routing and check RMII timing, oscillator total ±50 ppm budget, duty cycle and the PHY's 150 ps peak-to-peak jitter requirement. The oscillator drives only the buffer input; check that load against its 15 pF rating. The fan-out's low output skew avoids directly combining all three receiver loads on the oscillator. Clock buffer, oscillator drawing.

The FPGA and Ethernet require 50 MHz clock signals but do not require two oscillators. The shared clock remains independent of FPGA gateware. FPGA **CCLK/P70** is a separate configuration clock. STM32 retains its **8 MHz crystal** for CPU/USB clocks and USB-only operation; it does not depend on the HP-powered 50 MHz source to boot.

3.2.3 FPGA configuration

Configuration status, straps and probe access

Provide accessible INIT_B and DONE probe pads with nearby GND. A suitable header pad can provide the same access. These signals help diagnose reset, configuration errors and completion, including passive checks in DEV mode.

Signal	Purpose and connection
INIT_B, P39	Active-low, open-drain initialization/configuration status. Pull up to +3V3_HP and provide probe access. Reserve its status function; no MCU GPIO connection.
DONE, P71	Configuration-completion indication to **STM32 PE3**, with FPGA-side pull-up and probe access. It does not establish design identity or application readiness.
HSWAPEN, P144	**Tie to GND** to enable weak user-I/O pull-ups during pre-configuration initialization and programming. Reserve the pin; never drive it from user logic.
SUSPEND, P73	**Tie directly to GND**; suspend mode is unused. Reserve the pin.

Behavior before configuration and during reset

FPGA state	User-I/O behavior
Supplies absent or ramping below operating thresholds	Do not rely on internal bias; external circuits must tolerate the power sequence.
Configuration reset / PROGRAM_B low, with valid rails	User output drivers are disabled; HSWAPEN low enables weak pull-ups during initialization.
Bitstream loading	Unused user I/O remains weakly pulled high; active configuration pins perform their configuration functions.
Startup complete	The bitstream controls output enables and pull settings; HSWAPEN does not impose a permanent pull-up policy.
STM32-only reset button	FPGA configuration and its configured I/O behavior continue.

HSWAPEN low supplies useful default highs, but its weak pulls do not replace the **220 Ω receiver pull-ups**, **2 kΩ HP driver-input pull-ups**, or defined post-configuration FSMC/flash bias. Keep external bias where electrical loading or inactive-state requirements need it. UG380, table 5-2 and initialization.

Set **M1/P60 low and M0/P69 high** for Master SPI (01). Keep an external flash-CS pull-up and define unused-pin behavior in every bitstream, including the programming bridge. Startup pull-ups must be included when checking expansion loads and MCU/FPGA power-domain isolation.

3.2.3.1 SPI NOR flash

Use **GD25Q16ESIG, C2693492**, **2.7–3.6 V, 16 Mbit / 2 MiB**, for x1 SPI configuration, in the **208 mil SOP-8** footprint. Validate power-on boot and the programming path with this exact device.

Its JEDEC ID is **C8 40 15** and QE defaults low. The programming tool needs explicit GigaDevice identification/status-register handling and must preserve the required /WP and /HOLD behavior. Implement and test this support before use. GigaDevice data sheet.

Function	FPGA package pin	Flash pin / connection
CSO_B / flash select	P38	1, /CS
CCLK / SPI clock	P70	6, CLK
DIN / flash-to-FPGA data	P65	2, DO/IO1
MOSI / FPGA-to-flash data	P64	5, DI/IO0
Master SPI strap M1	P60	Pull low
Master SPI strap M0	P69	Pull high
Configuration reset PROGRAM_B	P37	Conditioned PON reset, 3.2.7
Configuration done DONE	P71	STM32 PE3 input; probe access

Initial configuration remains **SPI x1**, starting at flash address zero. The existing implementation uses CCLK startup and a 2 MHz initial configuration clock. Keep /CS inactive while its driver is released. For the new board, provide defined inactive levels for flash /WP/IO2 and /HOLD/IO3. Any future quad wiring needs its own pin review. See the existing flash workflow. Fit resistor pull-ups on /CS, /WP and /HOLD; check the GD power-up timing before releasing configuration reset.

3.2.3.2 JTAG to STM32 and local 14-pin connector

Signal	STM32 GPIO	STM32 LQFP100 pad	FPGA TQG144 pad	MCU idle / DEV	MCU programming
TCK	PC0	15	P109	Input, no pulls	Output
TMS	**PE2**	**1**	P107	Input, no pulls	Output
TDI	PC2	17	P110	Input, no pulls	Output
TDO	PC3	18	P106	Input, no pulls	Input
DONE	**PE3**	**2**	P71	Input, no pulls	Input

The STM32 uses GPIO to control FPGA JTAG; this is independent of its own SWD debug port. Keep MCU JTAG, SWO and parallel trace disabled. PA13/PA14 remain dedicated to SWD; PA15/PB3/PB4 are expansion GPIO.

Connect a local **14-pin FPGA JTAG connector**, as requested, in parallel with the MCU branches. Use the standard keyed **2 × 7, 2.54 mm Platform Cable** interface, with these electrical assignments:

Header pin(s)	Connection
1, 3, 5, 7, 9, 11	GND
2	+3V3_HP target VREF; not an input power source for the board
4	TMS, FPGA P107
6	TCK, FPGA P109
8	TDO, FPGA P106
10	TDI, FPGA P110
12	NC
13	Cable PGND/pseudo-ground function; leave NC
14	Cable HALT function in JTAG mode; leave NC

Check the user's dongle orientation, pin-1 marking and footprint numbering against the actual 2.54 mm cable. These signal assignments follow DS593 figure 14; the accepted header is TME **CONNFLY DS1013-14SSIBG / ZL231-14PG**; verify its drawing, height and mated-cable clearance before capture. Do not wire INIT_B/PROGRAM_B to the cable's alternate serial/SPI-mode positions.

100–220 Ω series footprints go in the three MCU output branches, not in a way that isolates the dongle from the FPGA. Bias the FPGA-side JTAG nets deliberately; MCU idle settings remain input/no pull. Series resistors do not replace programmer ownership: fit DEV before external programming.

DONE's external pull-up belongs on +3V3_HP; PE3 must not backfeed the FPGA from the MCU rail. Claim JTAG only with HP power present and DEV absent. The DEV contract is specified in **3.3.6**. JPROGRAM supports FPGA reload without pulsing the shared physical PROGRAM_B/MCU reset path.

3.2.4 FSMC interface to STM32 and wire-wrap area

FSMC signal	STM32 GPIOs, low bit first	FPGA pads, low bit first
AD0-AD3	PD14, PD15, PD0, PD1	P100, P99, P98, P97
AD4-AD12	PE7, PE8, PE9, PE10, PE11, PE12, PE13, PE14, PE15	P95, P94, P93, P92, P88, P87, P85, P84, P83
AD13-AD15	PD8, PD9, PD10	P82, P81, P80
NOE	PD4	P79
NWE	PD5	P78
NADV / FSMC_NL	PB7	P75
AUX2, fitted; application-defined	PB9	P74
AUX1, fitted; application-defined	PB8	P67

These are 19 active transport wires plus **two fitted bidirectional auxiliary nets, AUX1 and AUX2**. Their function is application-defined; neither is assigned to IRQ or DRQ. Keep both MCU/FPGA ends as inputs until assigned. No physical NE signal is used by the current single-FSMC-device implementation. Keep control strobes externally inactive during reset; release the AD bus outside FPGA read windows. Preserve the documented register protocol initially. FSMC implementation and timing

Agreed FSMC release and FPGA-only applications

Route access to **all 19 FSMC nets**, plus fitted AUX1/P67 and AUX2/P74, to the expansion area. They remain connected to the STM32; reuse requires its pins to be in high impedance, not merely an idle FSMC bus. For an application that does not use the STM32 backend, the loader and application must:

1. Select the released-bus policy **before** configuring an FPGA design that drives those nets. Do not briefly initialize FSMC first.
2. Stop transport accesses, DMA and interrupts, then disable FSMC operation. Disabling its clock alone is insufficient.
3. Return **every connected MCU pin**—AD0–AD15, NOE, NWE and NADV, plus PB8/PB9 auxiliary connections—to GPIO input with **no pull-up/down**, or an equivalent documented high-impedance mode. Release the strobes too.
4. Preserve that state through handoff, errors and resets. Before re-enabling transport, first load/verify an FPGA design that releases conflicting expansion outputs. Do not access the FSMC memory window in expansion mode.

Fixed external bias remains a load when GPIOs are released. Use weak FPGA-side control pull-ups and omit unnecessary AD pull-ups, with configurable/ DNP footprints where reuse needs flexibility. Provide disconnect-link footprints at the MCU branches if practical; expansion posts belong on the **FPGA side** of those links.

A high-impedance MCU is still electrically connected: expansion signals must stay within both devices' pin ratings, including powered-off conditions. These are **3.3 V logic nets**; external drivers/receivers handle other levels. If the MCU is omitted or can be unpowered while expansion drives these nets, open the links or provide isolation. GPIO/FSMC reference.

3.2.5 HP 21xx I/O pins connected to the FPGA

Use HP21xxReceiverDriver as the circuit reference. The upstream revision inspected for this review is `fc92e9d4aa22cc9dab332444d343fde6f802ec13` (retrieved 2026-10-31). Its schematic SHA-256 is `aa0e1389faad65e0f919ea2b79baefb2bf7e7c6236e3b779a07f916a1f034ae4`.

The exported netlist confirms **R49–R54 are 2 kΩ pull-ups to `PLUS3_3V_IN`**, supplied on driver-module J1 pin 1. The upstream README notes that the updated circuit still needs hardware validation.

Integrate the functions into the new board, adapting module power distribution to the central supplies. Repeated regulator/connector blocks need not be copied literally. Retain the actual electrical topology unless a change is reviewed: 74ACT04-driven transistor emitter outputs and transistor receivers, with the source's MMBT2369A/diode/resistor networks.

Both receiver and driver invert voltage. A high FPGA driver-input level is the inactive HP output condition. Provide external **3.3 V** driver-input bias so outputs remain inactive before configuration and during JTAG bridge loading. Use **2 kΩ** for these driver-input pull-ups and verify loading and inactive levels in the

completed circuit. The power review also identifies about **0.49 A direct +5 V driver base-bias current** with all 24 drivers inactive. Those 220 Ω resistors can dissipate about 90 mW each; select appropriate packages/derating rather than copying every 0402 footprint.

HP backplane signals are emitter-follower driven with pulls toward -2 V; neither the inactive low nor the active high is an ordinary push-pull 3.3 V logic output. Account for receiver current, termination and propagation delay. HP interfacing guide, sections 5-8/5-9

Provide **43 receiver channels and 24 driver channels**, including separate RUN receive/drive paths. Appendix A and the pin table give the allocation.

The initial timing reference remains the user-reported 200 ns shortest HP pulse, ten cycles at 50 MHz. Validate actual thresholds, load, propagation and setup/hold on hardware. No generalized HP-model timing guarantee is claimed.

3.2.6 Uncommitted pins and wire-wrap area

Agreed: provide a small plated-through-hole prototyping area accepting proper wire-wrap posts, with a top connector and labelled access to uncommitted FPGA I/O, including **P59, P61, P62 and P66**. These are general-purpose expansion nets. Route the reusable FSMC pins above to the same area.

Expose ground and labelled logic supplies with room for small drivers and receivers. Match the supplied prototype wire-wrap geometry: **2.54 mm nominal grid, 1.6 mm circular copper pads and 0.8 mm plated drills** on both copper layers. These values were read from the prototype's 5,379 isolated single-pad footprints. Confirm finished-hole fabrication tolerance against the actual wire-wrap posts; keep the specified nominal pad/drill geometry.

Group	FPGA pins / access	Use
Uncommitted GPIO	**P59, P61, P62, P66**	General experimentation
Fitted auxiliary interconnect	**P67 ↔ PB8 AUX1; P74 ↔ PB9 AUX2**	Application-defined function/direction; access requires both devices released
Active FSMC	**19 pins in 3.2.4**	Reusable under the complete transport-release policy

Route access to **25 FPGA signal nets**, including reusable transport and AUX1/AUX2, plus supply/ground and the spare RS-232 channel nodes in 3.3.5. Route the **25 uncommitted/conditional STM32 pins** listed in 3.3.9, including PA4/PA5. AUX1/AUX2 are shared MCU/FPGA nets; their two ends are not independent expansion signals. Keep expansion stubs short and check FSMC loading.

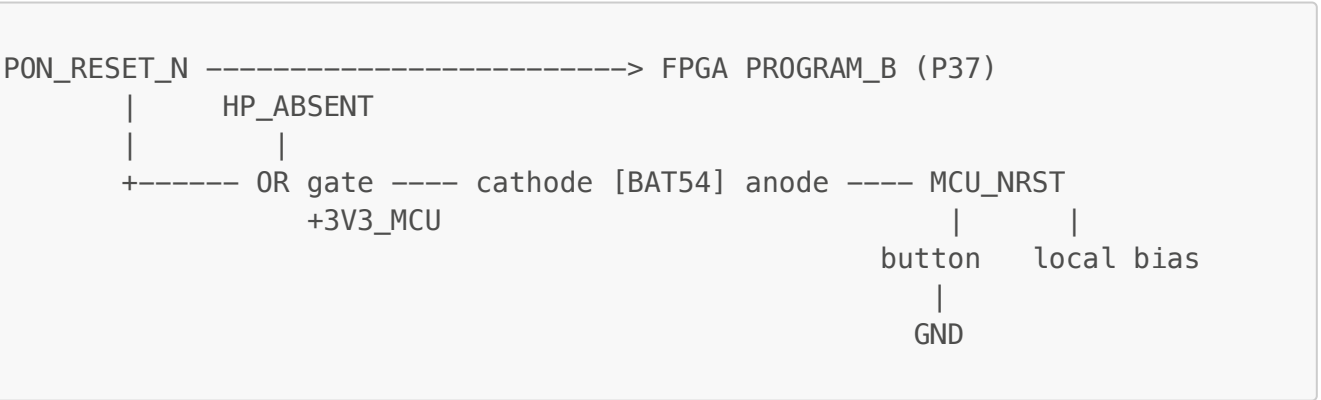
Informative analog capability: PA4 and PA5 are ordinary uncommitted expansion pins that also support the STM32's two native **12-bit DACs**. They can be used for analog or digital experiments, including an X/Y scope example. The board does not dedicate them to a DAC application or require X/Y buffers/connectors. Observe ST's voltage/load limits and add external conditioning in the expansion area as the experiment requires. STM32 DAC/pin characteristics.

Area dimensions, numbered external/top-connector allocation and post part remain layout choices. The spare RS-232 line side reaches this external I/O connector; keep it distinct from 3.3 V GPIO posts and HP backplane signals.

3.2.7 Reset

Condition HP PON to 3.3 V with the resistor/diode network in IMG_7708.JPG. Connect it to FPGA PROGRAM_B and to the power-aware, diode-isolated MCU reset branch. Do not connect raw negative PON to either device's reset pin.

Element	Connection
R_PON	470 Ω between HP_PON_RAW and PON_SER
D_PON, 1N4148	Anode PON_RESET_N, cathode PON_SER
R_RESET	1 kΩ from +3V3_HP to PON_RESET_N
R_BIAS	470 Ω from +3V3_HP to RESET_CLAMP_REF
D_REF, 1N4148	Anode RESET_CLAMP_REF, cathode GND
D_CLAMP, 1N4148	Anode RESET_CLAMP_REF, cathode PON_RESET_N
FPGA PROGRAM_B	P37 directly to PON_RESET_N
MCU reset gate	**TI SN74LVC1G32DBVR**, MCU_PON_RELEASE = PON_RESET_N OR HP_ABSENT, powered by +3V3_MCU; 100 nF local bypass
D_MCU, BAT54	**Anode MCU_NRST, cathode MCU_PON_RELEASE**
STM32 reset button	MCU_NRST to GND when pressed
MCU local bias	10 kΩ to +3V3_MCU, 100 nF to GND at MCU_NRST



Event	Required electrical behavior
HP PON reset with HP power present	Reset MCU and clear FPGA configuration.
MCU reset button / SWD / MCU-generated NRST	Reset MCU without pulling FPGA PROGRAM_B low.
JTAG JPROGRAM	Clear/reconfigure FPGA internally without asserting physical PROGRAM_B.

Event	Required electrical behavior
HP power absent, MCU USB-powered	HP reset branch releases MCU; FPGA/HP circuitry stays unpowered.

Use the PA0 HP-presence detector as a hardware gate input, independent of firmware. The gate supports partial-power operation; verify total reset-low voltage, including diode drop, and release timing. The circuit has shown repeated successful prototype starts, but voltage and startup margins need measurement. The –2 V rail alone does not establish the loaded PON level because the conditioner supplies current into the bus pull. See 3.3.7 and open issue H04.

3.2.8 Complete FPGA package-pin inventory

This table accounts for **all 144 TQG144 pins**, using the installed Xilinx `xc6slx9_tqg144.bsd`, the workbooks and the current controller pin-map reader. The HP and 19-wire transport assignments are listed explicitly below. TMS connects to STM32 PE2; DONE connects to STM32 PE3. **Usage** groups pins by their board function. Direction is relative to the FPGA. `Conditional` means a connected MCU or other fixed function must first be released/disconnected. Supply/configuration pins are accounted for explicitly and are not spare GPIO.

Machine-readable allocation and detailed notes: FPGA pin inventory.

Pad	Pin	Usage	Function / connection	Direction	Expansion
1	IO_P1	HP-BUS	HP IOB2 TX; HP contact 41	O	No
2	IO_P2	HP-BUS	HP IOB1 TX; HP contact 38	O	No
3	GND	POWER	GND	Power	No
4	VCCO_3	POWER	+3V3_HP	Power	No
5	IO_P5	HP-BUS	HP SCM_H; HP contact 37	I	No
6	IO_P6	HP-BUS	HP IOB0 TX; HP contact 35	O	No
7	IO_P7	HP-BUS	HP SCL_H; HP contact 34	I	No
8	IO_P8	HP-BUS	HP IRQH; HP contact 33	O	No
9	IO_P9	HP-BUS	HP SIR; HP contact 32	I	No
10	IO_P10	HP-BUS	HP IOB10 RX; HP contact 31	I	No
11	IO_P11	HP-BUS	HP IOB2 RX; HP contact 30	I	No
12	IO_P12	HP-BUS	HP IOB1 RX; HP contact 29	I	No
13	GND	POWER	GND	Power	No
14	IO_P14	HP-BUS	HP IOB9 RX; HP contact 28	I	No
15	IO_P15	HP-BUS	HP IOB8 RX; HP contact 27	I	No
16	IO_P16	HP-BUS	HP IOB0 RX; HP contact 26	I	No

Pad	Pin	Usage	Function / connection	Direction	Expansion
17	IO_P17	HP-BUS	HP SFS; HP contact 25	I	No
18	VCCO_3	POWER	+3V3_HP	Power	No
19	VCCINT	POWER	+1V2_FPGA	Power	No
20	VCCAUX	POWER	+3V3_HP	Power	No
21	IO_P21	HP-BUS	HP IOI; HP contact 24	I	No
22	IO_P22	HP-BUS	HP PRH; HP contact 23	I	No
23	IO_P23	HP-BUS	HP STC; HP contact 22	I	No
24	IO_P24	HP-BUS	HP CLC; HP contact 21	I	No
25	GND	POWER	GND	Power	No
26	IO_P26	HP-BUS	HP IOO; HP contact 20	I	No
27	IO_P27	HP-BUS	HP SRQ; HP contact 19	O	No
28	VCCINT	POWER	+1V2_FPGA	Power	No
29	IO_P29	HP-BUS	HP IOB16_OR_BIOS_N; HP contact 18	I	No
30	IO_P30	HP-BUS	HP POPIO; HP contact 17	I	No
31	VCCO_3	POWER	+3V3_HP	Power	No
32	IO_P32	HP-BUS	HP SCL_L; HP contact 16	I	No
33	IO_P33	HP-BUS	HP IOG; HP contact 15	I	No
34	IO_P34	HP-BUS	HP SCM_L; HP contact 14	I	No
35	IO_P35	HP-BUS	HP CRS; HP contact 13	I	No
36	VCCAUX	POWER	+3V3_HP	Power	No
37	PROGRAM_B	CONFIGURATION	Conditioned PON reset	I	No
38	CSO_B	CONFIGURATION	Flash /CS pin1	O	No
39	INIT_B	CONFIGURATION	Configuration status, FPGA-rail pull-up and probe access	I / open-drain O	No
40	IO_P40	HP-BUS	HP SKF; HP contact 12	O	No
41	IO_P41	HP-BUS	HP T3; HP contact 11	I	No
42	VCCO_2	POWER	+3V3_HP	Power	No
43	IO_P43	HP-BUS	HP IAK; HP contact 10	I	No
44	IO_P44	HP-BUS	HP STF; HP contact 9	I	No

Pad	Pin	Usage	Function / connection	Direction	Expansion
45	IO_P45	HP-BUS	HP IEN; HP contact 8	I	No
46	IO_P46	HP-BUS	HP CLF; HP contact 7	I	No
47	IO_P47	HP-BUS	HP IRQL; HP contact 6	O	No
48	IO_P48	HP-BUS	HP SFC; HP contact 5	I	No
49	GND	POWER	GND	Power	No
50	IO_P50	HP-BUS	HP FLGL; HP contact 4	O	No
51	IO_P51	HP-BUS	HP PRL; HP contact 3	O	No
52	VCCINT	POWER	+1V2_FPGA	Power	No
53	VCCAUX	POWER	+3V3_HP	Power	No
54	GND	POWER	GND	Power	No
55	IO_P55	HP-BUS	HP IOB9 TX; HP contact 56	O	No
56	IO_P56	OTHER	Shared 50MHz reference from CDCLVC1104 Y0	I	No
57	IO_P57	HP-BUS	HP IOB4 TX; HP contact 42	O	No
58	IO_P58	HP-BUS	HP IOB12 TX; HP contact 57	O	No
59	IO_P59	UNCOMMITTED	Uncommitted GPIO	I/O	Yes
60	M1	CONFIGURATION	Master SPI strap low	I	No
61	IO_P61	UNCOMMITTED	Uncommitted GPIO	I/O	Yes
62	IO_P62	UNCOMMITTED	Uncommitted GPIO	I/O	Yes
63	VCCO_2	POWER	+3V3_HP	Power	No
64	MOSI	CONFIGURATION	Flash DI/IO0 pin5	O	No
65	DIN	CONFIGURATION	Flash DO/IO1 pin2	I	No
66	IO_P66	UNCOMMITTED	Uncommitted GPIO	I/O	Yes
67	IO_P67	FPGA-CONTROL	AUX1 ↔ STM32 PB8	I/O	Conditional
68	GND	POWER	GND	Power	No
69	M0	CONFIGURATION	Master SPI strap high	I	No
70	CCLK	CONFIGURATION	Flash CLK pin6	O	No
71	DONE	CONFIGURATION	STM32 PE3; FPGA-side pull-up	I/O	No
72	CMPCS_B	CONFIGURATION	Reserved: leave NC or vendor-approved bias	Reserved	No

Pad	Pin	Usage	Function / connection	Direction	Expansion
73	SUSPEND	CONFIGURATION	Tie GND when unused	I	No
74	IO_P74	FPGA-CONTROL	AUX2 ↔ STM32 PB9	I/O	Conditional
75	IO_P75	FPGA-FSMC	FSMC_NADV ↔ STM32 PB7	I	Conditional
76	VCCO_1	POWER	+3V3_HP	Power	No
77	GND	POWER	GND	Power	No
78	IO_P78	FPGA-FSMC	FSMC_NWE ↔ STM32 PD5	I	Conditional
79	IO_P79	FPGA-FSMC	FSMC_NOE ↔ STM32 PD4	I	Conditional
80	IO_P80	FPGA-FSMC	FSMC_AD15 ↔ STM32 PD10	I/O	Conditional
81	IO_P81	FPGA-FSMC	FSMC_AD14 ↔ STM32 PD9	I/O	Conditional
82	IO_P82	FPGA-FSMC	FSMC_AD13 ↔ STM32 PD8	I/O	Conditional
83	IO_P83	FPGA-FSMC	FSMC_AD12 ↔ STM32 PE15	I/O	Conditional
84	IO_P84	FPGA-FSMC	FSMC_AD11 ↔ STM32 PE14	I/O	Conditional
85	IO_P85	FPGA-FSMC	FSMC_AD10 ↔ STM32 PE13	I/O	Conditional
86	VCCO_1	POWER	+3V3_HP	Power	No
87	IO_P87	FPGA-FSMC	FSMC_AD9 ↔ STM32 PE12	I/O	Conditional
88	IO_P88	FPGA-FSMC	FSMC_AD8 ↔ STM32 PE11	I/O	Conditional
89	VCCINT	POWER	+1V2_FPGA	Power	No
90	VCCAUX	POWER	+3V3_HP	Power	No
91	GND	POWER	GND	Power	No
92	IO_P92	FPGA-FSMC	FSMC_AD7 ↔ STM32 PE10	I/O	Conditional
93	IO_P93	FPGA-FSMC	FSMC_AD6 ↔ STM32 PE9	I/O	Conditional
94	IO_P94	FPGA-FSMC	FSMC_AD5 ↔ STM32 PE8	I/O	Conditional
95	IO_P95	FPGA-FSMC	FSMC_AD4 ↔ STM32 PE7	I/O	Conditional
96	GND	POWER	GND	Power	No
97	IO_P97	FPGA-FSMC	FSMC_AD3 ↔ STM32 PD1	I/O	Conditional
98	IO_P98	FPGA-FSMC	FSMC_AD2 ↔ STM32 PD0	I/O	Conditional
99	IO_P99	FPGA-FSMC	FSMC_AD1 ↔ STM32 PD15	I/O	Conditional
100	IO_P100	FPGA-FSMC	FSMC_AD0 ↔ STM32 PD14	I/O	Conditional
101	IO_P101	HP-BUS	HP IOB7 RX; HP contact 84	I	No
102	IO_P102	HP-BUS	HP IOB15 RX; HP contact 83	I	No

Pad	Pin	Usage	Function / connection	Direction	Expansion
103	VCCO_1	POWER	+3V3_HP	Power	No
104	IO_P104	HP-BUS	HP IOB14 RX; HP contact 82	I	No
105	IO_P105	HP-BUS	HP IOB6 RX; HP contact 81	I	No
106	TDO	JTAG	STM32 PC3 and 14-pin header pin8	O	No
107	TMS	JTAG	STM32 PE2; header pin4	I	No
108	GND	POWER	GND	Power	No
109	TCK	JTAG	STM32 PC0 and header pin6	I	No
110	TDI	JTAG	STM32 PC2 and header pin10	I	No
111	IO_P111	HP-BUS	HP IOB5 RX; HP contact 80	I	No
112	IO_P112	HP-BUS	HP IOB13 RX; HP contact 79	I	No
113	GND	POWER	GND	Power	No
114	IO_P114	HP-BUS	HP IOB12 RX; HP contact 78	I	No
115	IO_P115	HP-BUS	HP IOB4 RX; HP contact 77	I	No
116	IO_P116	HP-BUS	HP IOB15 TX; HP contact 74	O	No
117	IO_P117	HP-BUS	HP SFSB_OR_BIOI_N; HP contact 73	I	No
118	IO_P118	HP-BUS	HP BIOO_N; HP contact 67	I	No
119	IO_P119	HP-BUS	HP PON; HP contact 66	I	No
120	IO_P120	HP-BUS	HP IOB14 TX; HP contact 65	O	No
121	IO_P121	HP-BUS	HP IOB3 RX; HP contact 64	I	No
122	VCCO_0	POWER	+3V3_HP	Power	No
123	IO_P123	HP-BUS	HP EDT; HP contact 62	I	No
124	IO_P124	HP-BUS	HP IOB13 TX; HP contact 61	O	No
125	VCCO_0	POWER	+3V3_HP	Power	No
126	IO_P126	HP-BUS	HP IOB11 RX; HP contact 60	I	No
127	IO_P127	HP-BUS	HP IOB10 TX; HP contact 58	O	No
128	VCCINT	POWER	+1V2_FPGA	Power	No
129	VCCAUX	POWER	+3V3_HP	Power	No
130	GND	POWER	GND	Power	No

Pad	Pin	Usage	Function / connection	Direction	Expansion
131	IO_P131	HP-BUS	HP IOB11 TX; HP contact 55	O	No
132	IO_P132	HP-BUS	HP IOB8 TX; HP contact 54	O	No
133	IO_P133	HP-BUS	HP IOB6 TX; HP contact 53	O	No
134	IO_P134	HP-BUS	HP IOB7 TX; HP contact 52	O	No
135	VCCO_0	POWER	+3V3_HP	Power	No
136	GND	POWER	GND	Power	No
137	IO_P137	HP-BUS	HP IOB5 TX; HP contact 51	O	No
138	IO_P138	HP-BUS	HP RUN_DRIVE; HP contact 50	O	No
139	IO_P139	HP-BUS	HP RUN; HP contact 50	I	No
140	IO_P140	HP-BUS	HP FLGH; HP contact 49	O	No
141	IO_P141	HP-BUS	HP ENF; HP contact 46	I	No
142	IO_P142	HP-BUS	HP IOB3 TX; HP contact 45	O	No
143	IO_P143	OTHER	Active-high FPGA LED	O	No
144	HSWAPEN	CONFIGURATION	Tie GND; enable pre-configuration weak pull-ups	I	No

3.3 STM32 — selected chip

Selected **STMicroelectronics STM32F407VGT6, LCSC C12345**, LQFP100, **1 MiB flash**. The pin inventory includes Ethernet, FPGA FSMC/JTAG/DONE, SDIO, USB, serial, SWD and board-control signals. PA9/PA10 serve USART1. Linker/startup definitions must target this flash density and application layout.

Implement supply pins, VDDA/VREF filtering, VCAP capacitors, VBAT, boot straps and local decoupling from the data sheet and hardware guide.

3.3.1 Power — USB-C / HP bus, Schottky OR and 3.3 V regulator

Use **two power Schottky diodes and an independent 3.3 V regulator**, as requested. One anode connects to HP 4.85 V, the other to USB VBUS; joined cathodes feed `MCU_VIN`. Use **MDD SS14/C2480**, SMA, 1 A / 40 V, with forward-drop/leakage/thermal checks against the operating allowance. BAT54 is a reset-signal diode, not the choice for these power paths.

```
HP 4.85 V ----->|-----+
USB VBUS ------>|-----+----- MCU_VIN ---- buck ---- +3V3_MCU
```

Use **TLV62569DBVR/C141836**, set to 3.3 V. A 1 A regulator covers the present allowance; using the same **2 A** part as the FPGA rails simplifies sourcing. Keep this output separate from `+3V3_HP`.

MCU-domain load	Planning allowance
STM32 operating current	**150 mA**
SD card operating/transient allowance	**300 mA**, verify the actual card
LED, bias and other support	**50 mA**
Total at 3.3 V	**500 mA**

These are design allowances, not measured maximums; recheck the MCU allowance with Ethernet MAC/DMA active. The PHY stays on +3V3_HP, so it does not add its load to USB-only recovery. At 4.50 V after an assumed 0.35 V diode drop and 90% regulator efficiency, the input is about **407 mA**; diode loss is about **0.14 W**. At a low 4.4 V USB input and 0.4 V diode drop, 4.0 V remains, above the regulator's input minimum. Review the full cable/protection/diode worst case. Detailed power calculation.

USB powers the MCU/SD recovery domain only; FPGA, HP output drivers and pendant remain HP-powered. Respect USB current permission, pre-enumeration/inrush and suspend limits; the regulator rating is not permission to draw 2 A from USB. The higher source supplies the diode OR, so USB can feed the MCU even while HP power is present. SD shares this rail to allow bench SD recovery.

The diode OR prevents supply backfeed but does not isolate GPIO/reset paths. In USB-only operation leave MCU-to-FPGA FSMC/JTAG outputs undriven, put DONE bias on the FPGA rail, and prevent other signals from energizing that rail. Detect HP power presence separately from USB. The PON reset branch requires the power-aware gating in **3.3.7**.

Use **Samsung CL10A225KO8NNNC/C23630, 2.2 µF/16 V X5R, 0603**, one at each VCAP pin. Provide 100 nF at each VDD pin and a 4.7 µF local MCU bulk capacitor; **CC0603KRX7R9BB104/C14663** and **GRM21BR71A475KA73L/C86039** are the selected common bypass/bulk parts. Keep VCAP outputs local to the MCU. Implement VDDA/VREF filtering and local capacitors from the ST hardware guide.

3.3.2 Clocks — 8 MHz crystal

Use an **independent 8 MHz HSE crystal** with load capacitors selected for its specified load capacitance and PCB stray capacitance. MCU startup and ROM USB DFU must not depend on a configured FPGA or its oscillator.

Use **168 MHz CPU / 48 MHz USB** application clocks. ROM DFU configures its own clocks. The independent 8 MHz reference meets the documented bootloader HSE range and is preferred over higher-frequency choices for ROM clock detection. The BOM proposes **Lucki L538S080L11L/C5917268**, a passive 8 MHz crystal with **12 pF load and 50 Ω ESR**, plus two **18 pF C0G capacitors/C1647** as a starting point (assuming 3 pF combined stray capacitance). Validate crystal drive level, oscillator gain margin and startup over temperature; these support selections remain provisional. RTC crystal/battery support is open; no RTC requirement was agreed. ST hardware guide, bootloader clock requirements.

3.3.3 USB-C and USB protection

Use the user-selected **HRO TYPE-C-31-M-12/C165948** USB-C receptacle, supplied and soldered by **JLCPCB SMT assembly**, with the selected **USBLC6-2SC6/C21713975** protection array. Use the exact HRO footprint and retention details. This restores the original HRO selection and removes the GCT/TME

proposal. The protection code is cataloged as **GOODWORK**, not STMicroelectronics; use its own data sheet and pinout. Selected connector, selected protection part.

USB remains a USB 2.0 **full-speed device** connection, with PA11 as D- and PA12 as D+. Wire both plug orientations correctly. Use separate nominal **5.1 kOhm Rd** terminations on CC1 and CC2 and provide connector-side ESD protection. No USB PD or host function has been requested. ST USB-C guidance

PA9/PA10 serve USART1. VBUS presence is sensed on PB0, LQFP100 pad 35, using **1.8 kΩ from USB VBUS to PB0 and 3.3 kΩ from PB0 to GND**, both 1%. The sense point is upstream of the USB power-OR diode. **High means VBUS present**. At 5.0 V, PB0 sees `5 × 3.3 / (1.8 + 3.3) = 3.235 V`, with about **0.98 mA** divider current. Including resistor tolerance, the node is approximately **2.827 V at 4.4 V VBUS** to **3.584 V at 5.5 V VBUS**.

PB0 is an **FT (5 V-tolerant) pin** in ST's pin table. Configure it as a **digital input without internal pulls**, and leave its ADC/analog function unused. The divider remains below the `VDD + 4 V` absolute limit even at VDD = 0, while providing a valid input high at the 3.3 V rail. Check leakage and connector transients in the schematic; do not treat the divider as ESD protection. ST pin/electrical limits, VBUS divider guidance.

Use a separate active-low HP-power detector on **PA0, pad 23**, because it also supplies a hardware `HP_ABSENT` signal for reset gating: NPN, 47 kΩ HP-to-base, 100 kΩ base-to-GND and 10 kΩ collector pull-up to `+3V3_MCU`.

PB0 provides application attach detection; it is separate from the peripheral's dedicated PA9 VBUS input. Configure the application USB stack appropriately and use PB0 for connection handling. Validate factory ROM DFU on the selected silicon revision with PA9/PA10 on USART1 and the pendant connected: the ROM's interface-selection code is independent of application sensing software. ST bootloader interfaces.

The Schottky-OR input powers only the separate MCU recovery domain from USB; it does not join VBUS to HP +5 V. Place ESD at the connector with short ground returns and connect its reference/VBUS pin per the selected GOODWORK diagram.

3.3.4 SD card

Use the four-bit SDIO transport. **Use PC7 for card present and PB1 for write protect.**

SD signal	STM32 GPIO / package pad
DAT0, DAT1, DAT2, DAT3	PC8/65, PC9/66, PC10/78, PC11/79
CLK	PC12/80
CMD	PD2/83
Mechanical card-present switch	**PC7/64**
Mechanical write-protect switch	**PB1/36**
Supply/return	`+3V3_MCU` / GND

Use **XUNPU SD-106M/C266603**, full-size push-push, 2.9 mm high. Its drawing has pad 10 CD and pad 11 WP switching to ground. Fit 10 kΩ pull-ups to the MCU rail and debounce in software. Holder drawing.

Condition	PC7 card present	PB1 write-protect	Meaning
Absent	High	High	No card; ignore WP
Inserted, locked	Low	High	Treat SD as read-only
Inserted, unlocked	Low	Low	SD writes permitted

Firmware must enforce the advisory write-protect tab for SD disk-image, configuration and metadata writes. Reading firmware/bitfiles from a locked card is permitted. Mechanical CD is distinct from SD contact 1 CD/DAT3. Assign the footprint/shield/common switch pads from the actual holder drawing. Provide local decoupling, bus pull-ups and proper clock routing; access, ESD and optional switched card power remain to be finalized. Start SDIO at 4 MHz polling; validate higher speeds separately.

3.3.5 Debug and pendant serial port — RS-232

Use **Molex 95501-2441**, TME symbol **MX-95501-2441**, **4P4C** (commonly sold as RJ9/RJ10), right-angle through-hole with PCB retention and a horizontal cable exit. Its **11.5 mm nominal height** leaves **1.3 mm nominal headroom** below the 12.8 mm limit. Use its own footprint and verify maximum installed height, board thickness, numbered contact pinout and latch access. No LEDs or magnetics are present in this pendant connector. Selected jack, TME ordering page.

The earlier **TE Connectivity 1-1705949-1**, also 11.5 mm high, is superseded for this BOM by the TME-sourced Molex part. The BOM includes a proposed **TPS2553DBVR/C55266** high-side current-limited +5 V switch. The pendant load/cable budget and its ILIM resistor remain open; do not infer them from the connector's 1.5 A contact rating.

Alternative: TE Connectivity 215875-1, also right-angle through-hole 4P4C, **11.5 mm** high, for a 1.6 mm PCB; this version has no panel stops. Review its own footprint and packaging rather than assuming interchangeability. Alternative jack and drawing. The previous **Würth 615004143821**, **14.5 mm** high, exceeds the corrected limit. Würth drawing.

The four cable conductors are **+5 V, GND, RS-232 TX and RS-232 RX**. Use **USART1 PA9/TX and PA10/RX**, through **TI SN75185DWR/C2671047**, SOIC-20, powered by HP +4.85 V/+12 V/–12 V. Use the first driver/receiver for the pendant. **SN74LVC2G125DCUR/C21404**, on `+3V3_MCU`, provides pendant RX level conversion and controlled TX isolation; its enables default to disabled when HP power is absent. USB-only recovery uses USB/SWD.

All remaining SN75185 channels are fitted and exposed:

Channel	SN75185 logic pin	RS-232 line pin	Destination
Pendant TX, driver 1	DA1, pin 16	DY1, pin 5	PA9 path to 4P4C TX
Pendant RX, receiver 1	RY1, pin 19	RA1, pin 2	4P4C RX to PA10 path
Expansion TX2	DA2, pin 15	DY2, pin 6	Logic post to external/top I/O connector

Channel	SN75185 logic pin	RS-232 line pin	Destination
Expansion TX3	DA3, pin 13	DY3, pin 8	Logic post to external/top I/O connector
Expansion RX2	RY2, pin 18	RA2, pin 3	External/top connector to 3.3 V logic post
Expansion RX3	RY3, pin 17	RA3, pin 4	External/top connector to 3.3 V logic post
Expansion RX4	RY4, pin 14	RA4, pin 7	External/top connector to 3.3 V logic post
Expansion RX5	RY5, pin 12	RA5, pin 9	External/top connector to 3.3 V logic post

Use **SN74LVC125APWR/C7813**, powered by `+3V3_HP`, to translate the four spare 5 V receiver outputs to **3.3 V** before the logic posts. Bias spare transmitter logic inputs to UART idle with 10 kΩ to `+3V3_HP`; allow users to patch them from available MCU/FPGA GPIO. Define the buffer enables for rail startup and release. All RS-232 line-side signals also reach labelled posts and the **external/top HP I/O connector**, with final connector contacts to allocate. Do not connect bipolar RS-232 levels to the HP backplane logic or raw 3.3 V GPIO. Transceiver, receiver buffer.

Provide local bypassing at +5 V, +12 V and –12 V supply pins. Allow **50 mA on each ±12 V rail** for the three loaded drivers plus static current; verify actual loads/short-circuit dissipation and cable performance. The numbered 4P4C pinout, cable wiring, baud rate and protected pendant +5 V limit remain in H10. The connector's contact rating does not define the pendant budget.

3.3.6 BOOT0 mode and DEV mode

BOOT0 / factory ROM recovery

Preserve BOOT0 selection and a BOOT1/PB2 pull-down. BOOT0 high with BOOT1 low at reset selects ST's factory system-memory loader, which supports USB DFU on PA11/PA12. The independent 8 MHz crystal satisfies the documented clock range. ROM DFU can install the resident SD loader at `0x08000000` even when application flash is invalid. Boot with BOOT0 low afterwards. Retain SWD as another recovery path. ST bootloader documentation

Provide a labelled BOOT0 jumper and MCU reset button. BOOT0 and DEV are separate controls: BOOT0 selects the MCU boot source, DEV selects FPGA JTAG ownership. Verify ROM recovery with the pendant connected and both supported power sources.

DEV jumper / FPGA JTAG ownership

Pin: PC6, STM32 pad 63, with an external **10 kΩ pull-down** and a jumper to `+3V3_MCU`. Fitted jumper means DEV.

- Read DEV before any possible FPGA JTAG operation.

- With DEV fitted, never drive TCK, TMS or TDI: **no chain scan, IDCODE read, USERCODE read, DONE-over-JTAG read, flash probe, JPROGRAM or bridge download**.
- Leave **PC0, PE2, PC2 and PC3** as inputs without MCU pull resistors throughout boot, application handoff, normal operation and error/recovery paths.
- Reading the separate DONE input is passive and permitted.
- Pass DEV/ownership state to the application; its JTAG code must obey it too.
- With DEV absent, automatic startup checks/updates may briefly claim JTAG; release the MCU outputs afterwards. External development requires DEV fitted.
- DEV inhibits automatic FPGA management, but not physical PON reset. Whether DEV also inhibits automatic **STM32 application replacement** is open; the user has only specified the FPGA/JTAG restriction so far.

3.3.7 Reset

At **MCU_NRST**, fit **10 kΩ to** `+3V3_MCU`, **100 nF to GND**, the reset button to GND, and the ST-LINK NRST connection. The BOM proposes **UNI-ROYAL 0603WAF1002T5E/C25804** and **CC0603KRX7R9BB104/C14663** for the local resistor/capacitor. Their nominal RC product is 1 ms; the internal NRST pull-up and thresholds affect release. The **BAT54 anode is at MCU_NRST**; its cathode is on the conditioned PON reset branch, via the power-aware gate in 3.2.7.

A bare diode connection to the HP-powered PON node would hold the USB-powered MCU low when that node is unpowered; a pull-up/capacitor/button does not remove that DC clamp. Use **SN74LVC1G32DBVR** on `+3V3_MCU`, with `MCU_PON_RELEASE = PON_RESET_N OR HP_ABSENT`, ahead of the BAT54 cathode. The PA0 HP-presence detector supplies HP_ABSENT in hardware before firmware starts. FPGA PROGRAM_B remains directly on conditioned PON.

This retains PON reset with HP power present, allows standalone USB startup, and isolates the MCU reset button from FPGA PROGRAM_B. Verify low-level margins including gate/diode drops, power sequencing and flash-ready delay. OR gate and partial-power characteristics.

3.3.8 Ethernet — 100BASE-TX and internal RMII

External interface: RJ45 to a 10/100 Ethernet LAN, maximum 100 Mb/s; no Gigabit or PoE requirement. **Internal interface:** the F407 Ethernet MAC supports **MII/RMII**, not RGMII. Select **RMII**, which uses seven clock/data/ control wires plus MDC and MDIO. STM32 Ethernet capability.

Selected Ethernet components

Function	Part	Source	Implementation notes
RMII PHY	**Microchip LAN8720A-CP-TR**, 24-QFN 4 × 4 mm	JLCPCB **C45223**, Standard assembly	10/100 RMII; shared external 50 MHz clock.
External magnetics	**Pulse H1102NLT**, 16-pin SMT transformer/choke module	JLCPCB **C18584**	Separate 1:1 TX/RX isolation transformers and common-mode chokes; non-PoE; Pulse lists LAN8720A compatibility. Nominal height **5.969 mm**.

Function	Part	Source	Implementation notes
Unshielded RJ45	**Adam Tech MTJ-885X1**, right-angle THT with PCB snap retention	**TME MTJ-885X1**	Plastic housing; **no metallic shield, magnetics or LEDs**. Nominal height approximately **11.7 mm**, leaving **1.1 mm** nominal margin. Confirm drawing tolerances and orderability before PCB release.
External indicators	Two board-mounted LEDs and separate current-limit resistors	JLCPCB; accepted **KT-0603R/C2286** and **470 Ω/C23179**	Link/activity and speed/status indication; wire LED polarity around PHY strap requirements. Initial red LEDs reduce distinct parts; brightness/current remain to validate.
RMII reference clock	Shared source and CDCLVC1104 fan-out, 3.2.2	**C5203552 / C485518**	Separate buffered PHY/MAC clock branches.

The **MTJ-885X1** is the selected low-cost mechanical candidate, but TME showed no stock during sourcing research. Procurement remains open; do not assume immediate availability. A same-function alternative needs its own height, electrical category, retention and footprint checks. The **MTJ-889X1** is nominally about **12.3 mm** high and has less margin; it also showed no stock. TME jack, Adam Tech catalog, JLCPCB magnetics, Pulse product/compatibility reference, Pulse drawing.

A stocked alternative is **Molex 85503-5001 / TME MX-85503-5001**: unshielded Cat 5e, right-angle THT with PCB retention, no magnetics or LEDs, **11.5 mm nominal** height. It is recommended for procurement but remains a zero-quantity alternative pending user selection. Its footprint and maximum installed height need review. Additional checked candidates and exclusions are in the BOM sourcing notes.

The earlier **Pulse JXD1-1V17NL** fits the height limit but is **shielded** and has integrated magnetics/LEDs, so it is superseded by the unshielded jack plus external parts. The **JXD1-1V07NL** and **HR911105A** remain excluded for excessive installed height. The shielded recessed Taoglas alternative does not satisfy the new unshielded preference.

The expanded BOM separately enumerates magnetics, external LEDs, PHY bias, terminations, filtering, decoupling, hardware reset and protection candidates. These are initial support-circuit selections, to be reconciled against the completed schematic rather than treated as an already validated reference design.

RMII pin allocation

Signal	STM32 pin / pad	Direction at MCU
REF_CLK, 50 MHz	**PA1/24**	Input
MDIO	**PA2/25**	Bidirectional
MDC	**PC1/16**	Output

Signal	STM32 pin / pad	Direction at MCU
CRS_DV	**PA7/32**	Input
RXD0	**PC4/33**	Input
RXD1	**PC5/34**	Input
TX_EN	**PB11/48**	Output
TXD0	**PB12/51**	Output
TXD1	**PB13/52**	Output
PHY reset, extra GPIO	**PE5/4**	Open-drain output

Use AF11 for the nine MAC/management pins. Keep parallel trace disabled on PE2–PE5. The LQFP100 pin map also accommodates all 19 FSMC wires, SDIO, PC7/PB1 card switches, PA9/PA10 serial, USB and SWD.

Clock, supply, reset and line-side implementation

Use the PHY's **REF_CLK input mode**. Shared fan-out Y1 feeds PHY XTAL1/CLKIN and Y2 feeds STM32 PA1; keep PHY XTAL2 unused. Clock power and routing are defined in 3.2.2 and remain independent of gateware and MCU HSE. PHY data sheet.

Power PHY analog/I/O supplies from `+3V3_HP`, with appropriate filtering, local decoupling and internal-regulator output capacitors. Do not tie its internal VDDCR regulator output to FPGA 1.2 V. Allocate **200 mA** on the shared HP-side rail for PHY, magnetics drive, LED currents and margin; the data-sheet chip-current figures exclude magnetics/LEDs, and 10 Mb/s can consume more transmitter current. This is an engineering allowance requiring verification, not a guaranteed PHY maximum. It raises the rail estimate to **1.21 A including the separate shared-clock allowance**, within the 2 A stage. Recheck MCU current with MAC/DMA.

Provide hardware PHY power-on reset timing, then allow PE5 to assert reset; meet the vendor's 25 ms supply-to-release and 100 µs reset-pulse requirements. Set mode/address/regulator/clock straps deliberately, including **nINTSEL high for REF_CLK input mode**, and account for LED loads on strap pins. Fit MDIO bias and choose an MDC rate within the PHY limit. Poll link status over MDIO initially; a PHY-interrupt connection is optional.

Use the **external H1102NLT** TX/RX centre taps and the Microchip/Pulse reference circuits. Fit PHY-side terminations, bias/bypass capacitors and a separate cable common-mode termination network as required; these are no longer hidden inside a MagJack. There is **no RJ45 shield terminal**. Review the chassis/ESD reference, cable-side protection and isolation layout explicitly. Route short, controlled-impedance MDI pairs. The BOM proposes **TPS3808G33DBVR/C43698**, CT connected through **100 kΩ to VDD**, for hardware POR (300 ms typical, 180 ms minimum); PE5 can additionally assert the open-drain nRST net. This closes the support-part proposal without claiming electrical validation. During USB-only operation the PHY is off; release RMII/management/reset outputs and do not backfeed its rail. Ethernet remains on the **STM32 schematic sheet**, preserving the four-sheet plan.

3.3.9 Complete STM32 package-pin inventory and spare access

The table accounts for **all 100 LQFP100 pins**, including power, ground, VCAP, reset, boot and HSE. Pin numbering follows ST DS8626 and the KiCad `STM32F407V_E-G_Tx` symbol; peripheral assignments follow ST's AF tables. ST package/peripheral reference.

Usage groups pins by their board function. Direction is at the MCU; FPGA-JTAG outputs become inputs in DEV. The CSV inventory also records detailed notes.

With all **19 FSMC pins active** and **AUX1/AUX2 fitted**, there are **20 available GPIOs plus five conditional GPIOs**, totaling **25 uncommitted MCU pins**. PA4/PA5 belong to this pool and may serve analog or digital experiments. PA15/PB3/PB4 are available because debug is **SWD only**; configure them as GPIO early and keep JTAG/SWO/parallel trace disabled.

The pool is PA3, PA4, PA5, PA6, PA8, PA15; PB3, PB4, PB5, PB6, PB10, PB14, PB15; PC13, PC14, PC15; PD3, PD6, PD7, PD11, PD12, PD13; PE0, PE1, PE6. The **five conditional pins** are PC13–PC15 (drive/speed restrictions, no LSE crystal on PC14/15) and PE0/PE1 (optional FSMC byte enables). **PB8/PB9 are AUX1/AUX2**, excluded from this free pool. AUX and FSMC posts remain accessible only with both connected devices released.

Route labelled expansion access and observe reset/ROM boot states and per-pin voltage/load limits. Native DAC capability of PA4/PA5 is informative in 3.2.6; there is no dedicated X/Y hardware requirement.

Pad	Pin	Usage	Function / connection	Direction	Expansion
1	PE2	FPGA-JTAG	FPGA_TMS	O	No
2	PE3	FPGA-STATUS	FPGA_DONE	I	No
3	PE4	LED	MCU_LED_N	O	No
4	PE5	ETHERNET	ETH_PHY_RESET_N	O (open drain)	No
5	PE6	UNCOMMITTED	Spare GPIO	I/O	Yes
6	VBAT	POWER	Backup supply; tie to MCU rail if no battery	Power	No
7	PC13	UNCOMMITTED	Spare GPIO	I/O	Conditional
8	PC14	UNCOMMITTED	Spare GPIO	I/O	Conditional
9	PC15	UNCOMMITTED	Spare GPIO	I/O	Conditional
10	VSS	POWER	Ground	Power	No
11	VDD	POWER	+3V3_MCU; all pins connected and locally decoupled	Power	No
12	PH0	CLOCK	HSE_OSC_IN	Analog I	No
13	PH1	CLOCK	HSE_OSC_OUT	Analog O	No
14	NRST	RESET	MCU reset button and power-aware PON reset branch	I	No

Pad	Pin	Usage	Function / connection	Direction	Expansion
15	PC0	FPGA-JTAG	FPGA_TCK	O	No
16	PC1	ETHERNET	ETH_MDC	O	No
17	PC2	FPGA-JTAG	FPGA_TDI	O	No
18	PC3	FPGA-JTAG	FPGA_TDO	I	No
19	VDD	POWER	+3V3_MCU; all pins connected and locally decoupled	Power	No
20	VSSA	POWER	Analog ground	Power	No
21	VREF+	POWER	Analog reference; filtered MCU analog rail	Power	No
22	VDDA	POWER	Analog supply; filtered +3V3_MCU	Power	No
23	PA0	POWER-SENSE	HP_POWER_PRESENT_N	I	No
24	PA1	ETHERNET	ETH_RMII_REF_CLK	I	No
25	PA2	ETHERNET	ETH_MDIO	I/O	No
26	PA3	UNCOMMITTED	Spare GPIO	I/O	Yes
27	VSS	POWER	Ground	Power	No
28	VDD	POWER	+3V3_MCU; all pins connected and locally decoupled	Power	No
29	PA4	UNCOMMITTED	Uncommitted GPIO / DAC_OUT1	I/O / Analog O	Yes
30	PA5	UNCOMMITTED	Uncommitted GPIO / DAC_OUT2	I/O / Analog O	Yes
31	PA6	UNCOMMITTED	Spare GPIO	I/O	Yes
32	PA7	ETHERNET	ETH_RMII_CRS_DV	I	No
33	PC4	ETHERNET	ETH_RMII_RXD0	I	No
34	PC5	ETHERNET	ETH_RMII_RXD1	I	No
35	PB0	POWER-SENSE	USB_VBUS_PRESENT	I	No
36	PB1	SD-CARD	SD_WRITE_PROTECT	I	No
37	PB2	BOOT/DEV	BOOT1 strap low	I	No
38	PE7	FPGA-FSMC	FSMC_AD4	I/O	Conditional
39	PE8	FPGA-FSMC	FSMC_AD5	I/O	Conditional
40	PE9	FPGA-FSMC	FSMC_AD6	I/O	Conditional

Pad	Pin	Usage	Function / connection	Direction	Expansion
41	PE10	FPGA-FSMC	FSMC_AD7	I/O	Conditional
42	PE11	FPGA-FSMC	FSMC_AD8	I/O	Conditional
43	PE12	FPGA-FSMC	FSMC_AD9	I/O	Conditional
44	PE13	FPGA-FSMC	FSMC_AD10	I/O	Conditional
45	PE14	FPGA-FSMC	FSMC_AD11	I/O	Conditional
46	PE15	FPGA-FSMC	FSMC_AD12	I/O	Conditional
47	PB10	UNCOMMITTED	Spare GPIO	I/O	Yes
48	PB11	ETHERNET	ETH_RMII_TX_EN	O	No
49	VCAP_1	POWER	Internal regulator output; local capacitor only	Power	No
50	VDD	POWER	+3V3_MCU; all pins connected and locally decoupled	Power	No
51	PB12	ETHERNET	ETH_RMII_TXD0	O	No
52	PB13	ETHERNET	ETH_RMII_TXD1	O	No
53	PB14	UNCOMMITTED	Spare GPIO	I/O	Yes
54	PB15	UNCOMMITTED	Spare GPIO	I/O	Yes
55	PD8	FPGA-FSMC	FSMC_AD13	I/O	Conditional
56	PD9	FPGA-FSMC	FSMC_AD14	I/O	Conditional
57	PD10	FPGA-FSMC	FSMC_AD15	I/O	Conditional
58	PD11	UNCOMMITTED	Spare GPIO	I/O	Yes
59	PD12	UNCOMMITTED	Spare GPIO	I/O	Yes
60	PD13	UNCOMMITTED	Spare GPIO	I/O	Yes
61	PD14	FPGA-FSMC	FSMC_AD0	I/O	Conditional
62	PD15	FPGA-FSMC	FSMC_AD1	I/O	Conditional
63	PC6	BOOT/DEV	DEV_MODE	I	No
64	PC7	SD-CARD	SD_CARD_PRESENT_N	I	No
65	PC8	SD-CARD	SDIO_DAT0	I/O	No
66	PC9	SD-CARD	SDIO_DAT1	I/O	No
67	PA8	UNCOMMITTED	Spare GPIO	I/O	Yes
68	PA9	SERIAL	USART1_TX	O	No

Pad	Pin	Usage	Function / connection	Direction	Expansion
69	PA10	SERIAL	USART1_RX	I	No
70	PA11	USB	USB_FS_DM	I/O	No
71	PA12	USB	USB_FS_DP	I/O	No
72	PA13	SWD	SWDIO	I/O	No
73	VCAP_2	POWER	Internal regulator output; local capacitor only	Power	No
74	VSS	POWER	Ground	Power	No
75	VDD	POWER	+3V3_MCU; all pins connected and locally decoupled	Power	No
76	PA14	SWD	SWCLK	I	No
77	PA15	UNCOMMITTED	Spare GPIO	I/O	Yes
78	PC10	SD-CARD	SDIO_DAT2	I/O	No
79	PC11	SD-CARD	SDIO_DAT3	I/O	No
80	PC12	SD-CARD	SDIO_CK	O	No
81	PD0	FPGA-FSMC	FSMC_AD2	I/O	Conditional
82	PD1	FPGA-FSMC	FSMC_AD3	I/O	Conditional
83	PD2	SD-CARD	SDIO_CMD	I/O	No
84	PD3	UNCOMMITTED	Spare GPIO	I/O	Yes
85	PD4	FPGA-FSMC	FSMC_NOE	O	Conditional
86	PD5	FPGA-FSMC	FSMC_NWE	O	Conditional
87	PD6	UNCOMMITTED	Spare GPIO	I/O	Yes
88	PD7	UNCOMMITTED	Spare GPIO	I/O	Yes
89	PB3	UNCOMMITTED	Spare GPIO	I/O	Yes
90	PB4	UNCOMMITTED	Spare GPIO	I/O	Yes
91	PB5	UNCOMMITTED	Spare GPIO	I/O	Yes
92	PB6	UNCOMMITTED	Spare GPIO	I/O	Yes
93	PB7	FPGA-FSMC	FSMC_NADV	O	Conditional
94	BOOT0	BOOT/DEV	Boot selection jumper/pull-down; ROM recovery	I	No
95	PB8	FPGA-CONTROL	FPGA_AUX1	I/O	Conditional

Pad	Pin	Usage	Function / connection	Direction	Expansion
96	PB9	FPGA-CONTROL	FPGA_AUX2	I/O	Conditional
97	PE0	UNCOMMITTED	Spare GPIO	I/O	Conditional
98	PE1	UNCOMMITTED	Spare GPIO	I/O	Conditional
99	VSS	POWER	Ground	Power	No
100	VDD	POWER	+3V3_MCU; all pins connected and locally decoupled	Power	No

3.4 Other hardware

This section holds hardware material without a dedicated subsection in the requested outline. It remains here for the user's next allocation pass.

Mechanical and manufacturing details

Use IOB Proto Board.kicad_pcb and its project as the mechanical input for layout: **1.6 mm PCB thickness**, with an Edge.Cuts bounding envelope of **195.980 × 220.8135 mm** in the supplied orientation. Preserve the shaped outline, edge fingers and connector positioning rather than replacing it with a rectangle. Check final mechanical fit and close the small existing outline endpoint gaps (largest measured **0.0074 mm**) in the new layout. The source project remains a reference input.

Card pitch is **5/8 inch = 15.875 mm (approximately 15.9 mm)**. With a **1.6 mm PCB** and allowance for leads/solder joints protruding from the **adjacent card's solder side into this card's component space**, the maximum component-side connector/component height is **12.8 mm above this PCB's top surface**. This is the resulting height allowance, already accounting for those adjacent-card protrusions. Include housing, stand-offs and shield/EMI tabs in each component's maximum installed height, including tolerances. Check cable mating, latch access and front clearance separately.

Through-hole connectors are acceptable and may be hand-soldered after JLCPCB assembles the PCB. Connector selection is not restricted to SMT or the JLCPCB/LCSC assembly catalog. Select wire-wrap area size, mounting/keep-outs and edge plating from this mechanical baseline. Transfer exact component identities into KiCad fields; add designators/footprints and placement data after capture.

Regulator power indication

Fit a separate power LED for **+1V2_FPGA**, **+3V3_HP** and **+3V3_MCU**, independent of firmware. Use the accepted **KT-0603R/C2286** and one **470 Ω/C23179** series resistor per LED. Each 3.3 V LED connects to its own output and GND. The red LED cannot run directly from 1.2 V: the proposed core indicator uses +3V3_HP -> 470 Ω -> LED -> MMBT2369ALT1G collector, emitter to GND, base from +1V2_FPGA through **1 kΩ/C21190**, and **100 kΩ/C25803** base-to-GND. This requires both core and HP 3.3 V power; it indicates rail presence, not precision power-good. Validate transistor drive, startup and visibility. All three LEDs and support parts are counted separately in the BOM. The power allowances retain margin for these loads.

MCU indication and SWD support

Use an **active-low PE4 status LED**. The accepted parts are KT-0603R/C2286 and 470 Ω/C23179; validate brightness and GPIO current at capture. The MCU debug header is **ST-LINK 2 × 10, 2.54 mm, right angle**; the accepted header is **CONNFLY DS1013-20RSIB / TME ZL231-20KG**. Use its own drawing and footprint. Implement the standard 20-pin electrical allocation for **SWD only**:

Header pin(s)	Connection
1, 2	`+3V3_MCU` target-voltage reference
7	PA13 SWDIO
9	PA14 SWCLK
15	MCU_NRST, at the local RC/button node
3, 5	GND, as ST recommends for the unused JTAG wires in SWD mode
4, 6, 8, 10, 12, 14, 16, 18, 20	GND
11, 13, 17, 19	NC; no SWO or debugger power connection

MCU JTAG and SWO/parallel trace stay disabled. PA15/PB3/PB4 are expansion GPIO and have no connection to their JTAG cable positions. The FPGA has its separate **2 × 7, 2.54 mm** JTAG header. ST-LINK UM1075, right-angle header.

Remaining hardware work is listed in **section 5**.

4. Software

The resident loader selects and validates a compatible application/FPGA pair. The pendant shall support dynamic selection of the application and its matching gateware from SD. Selection and switchover must follow the same compatibility checks and ownership rules as startup. Hardware DEV, reset and power/transport ownership apply to both loader and application. This section specifies the platform firmware to implement.

4.1 Configuration

Use **one configuration file**, **`/HP21xx_IO_FPGA.cfg`**, shared by the resident loader and the selected application. Its `[profile]` section selects `name`; application-specific settings are in the section with that exact name. For `name=hp13210`, the application reads **`[hp13210]`** from the same file.

```
[profile]
name=hp13210
hardware=hp21xx-integrated
interface_version=1
transport=fsmc

[stm32]
application_id=0x13210001
build_id=hp13210-20261101-a
image=/firmware/hp13210.bin
image_size=<application byte count>
```

```
sha256=<64 hexadecimal digits>

[fpga]
image=/fpga/hp13210.bit
image_size=<bitfile byte count>
usercode=0x13210001
sha256=<64 hexadecimal digits>

[hp13210]
drive0=drive0.img
drive1=none
drive2=none
drive3=none
```

The example shows the schema; replace the size/hash placeholders with values from the built artifacts. Keep firmware, bitfiles and disk images as separate data files, while all platform/application settings live in this one `.cfg`.

- `[profile]`, `[stm32]` and `[fpga]` hold platform selection and image identity.
- `name` selects the application section **case-sensitively**. The loader validates platform sections; the application validates its named section. Other application sections may coexist without being interpreted or rejected by the active application's key parser.
- For `[hp13210]`, `drive0`–`drive3` are filenames relative to `/images`. Missing keys, empty values or `none` leave a drive offline. Images are opened read/write only when the card is present and writable.
- `transport=fsmc` enables the backend transport; `transport=none` leaves all MCU transport pins released for FPGA-only expansion use.
- Use a shared section-aware parser and reject duplicate keys/sections or a missing selected application section. Validate configuration and image files before erasing anything or enabling the HP interface.
- DEV is a physical override: no file setting can authorize FPGA JTAG in DEV.

Application ID identifies the application, build ID its version, and SHA-256 verifies the file contents. FPGA USERCODE identifies the running design/build; its file has a separate hash. Validate board target and interface version as well as IDs/hashes. Keep a validated snapshot of the selected configuration through loader-to-application handoff so both use the same profile.

4.2 Boot-loader

Use a resident loader at the normal STM32 flash boot address. Allocate the **1 MiB STM32F407VG** flash as follows:

Region	Address range	Capacity
Resident SD/platform loader	`0x08000000`-`0x0801FFFF`	128 KiB; sectors 0-4
Application and metadata	`0x08020000`-`0x080FFFFF`	**896 KiB**; sectors 5-11

Implement matching linker/startup-vector definitions and measure the final loader against its allocation. Keep the loader outside every normal application erase operation. The sector boundaries follow STM32

RM0090. Image validation and handoff are described below; final metadata allocation is still open.

Normal boot sequence:

1. Establish GPIO defaults with FSMC/JTAG released; read DEV and power presence before any possible FPGA access. Keep expansion/HP outputs in their defined startup states.
2. Mount SD with bounded retries; read/validate the selected profile and the files needed for an update before erasing existing images.
3. Compare the installed application with the expected hash; skip programming if it matches. Otherwise program only the application region, verify and commit its valid marker last.
4. With HP power present and DEV absent, check FPGA readiness and the running design identity; update SPI flash when required using a verified JTAG-to-SPI bridge. The bridge can be a support file on SD. Verify the flash image and reload from flash.
5. Require the selected firmware/FPGA interface agreement before enabling HP service; release JTAG and hand off cleanly to the application.

On interrupted application programming, the resident loader must still start and retry from SD. Validate installed images before executing them. A single application slot gives reinstall recovery, not preservation of the old image; A/B slots, rollback and a golden FPGA image are optional future extensions. Default no-card behavior is to boot an already valid installed application, otherwise wait in recovery. Exact recovery/error indication and last-profile metadata storage remain open. Avoid flash writes on every matching boot.

4.2.1 Loading application code and application IDs

The resident loader owns normal application installation. Compare the selected profile's **application ID, build ID, target, interface version, image length and SHA-256** against installed content. Skip erase/programming when the expected image is already valid and matches.

Use a linker-retained **application descriptor at a defined offset within the application image** containing its ID/build/target/interface/length. Keep the Cortex-M vector table at the application's linked base; do not prepend a header there without changing the link/vector design. The descriptor need not contain its own whole-file hash; the manifest supplies that, avoiding a self-referential hash. Exact descriptor offset/encoding and persistent valid- marker location remain to be designed.

For a replacement, validate the SD file completely before erasing, enforce card write-protect policy for any SD changes, erase **only application/metadata sectors**, program, read back and verify the expected hash/descriptor. Commit the valid marker last. Reject an application larger than its allocation or with invalid stack/reset-vector addresses. Keep the resident loader and its recovery route intact across a power interruption. A single slot recovers by reinstallation from SD; retaining a runnable old version requires another slot.

For handoff, quiesce loader interrupts/DMA/peripherals, establish the application's vector table/VTOR, stack and reset entry, and pass the profile, DEV/JTAG ownership, power state and FSMC-release contract. Application startup must preserve the relocated-vector arrangement. The 128 KiB loader allocation includes SD/filesystem/configuration and programming support; verify its final size fits the assigned sectors.

4.2.2 Updating FPGA SPI flash and bitfile ID

Use the Spartan-6 **32-bit USERCODE**, set by BitGen, for the expected running design tag. Example: `-g UserID:0x13210001`; JTAG USERCODE instruction is `001000`/`0x08`. Hardware IDCODE identifies the FPGA part, not its design. Assign unique non-default build IDs and retain full image hashes in profiles. Spartan-6 USERCODE

USERCODE identifies the running configuration, which could be a temporary SRAM download from the dongle. It alone does not prove persistent SPI contents. Verify flash when updating and check DONE/identity again after reloading from flash. How to establish persistent-image identity after an external programming session is an open policy item; a cached ID must not silently certify changed flash. Correct handling of the `.bit` header, flash byte order and startup clocks must match the existing verified programmer.

DONE can be read directly on **PE3**; JTAG can also report it in Capture-IR bit 5 or the STAT register bit 13. Reading through JTAG is forbidden in DEV mode. JPROGRAM (`0x0B`) supports internal reconfiguration without pulsing the shared physical PROGRAM_B reset node. Spartan-6 JTAG configuration/status

Update sequence: with HP power present and DEV absent, withdraw peripheral/backend readiness, validate the selected bitfile and target, then claim JTAG and load the SPI programming bridge. Identify the flash using its actual JEDEC ID and part-specific status/erase/program handling. Program and read back the intended flash bytes, then invoke internal reload from flash and verify DONE plus the expected USERCODE before returning to service. Release MCU JTAG outputs afterwards. Do not pulse physical PROGRAM_B from the MCU for this operation, since the shared reset path would also reset it.

A flash interruption can leave FPGA configuration invalid. Recovery through the independently booting MCU/SD or external JTAG must remain possible; a golden-image/rollback scheme is not yet selected. Honor the **PB1** SD write-protect input for any SD metadata updates. DEV mode skips **all** these JTAG operations, including otherwise harmless identity checks; software reports management as inhibited rather than attempting to repair the external developer's image.

4.3 Application program

The application implements the chosen peripheral's higher-level behavior. The initial disk application remains HP 13210/7900 emulation using the existing FSMC register/command protocol and SD disk images; another profile may supply a different application and matching FPGA image. Application-specific settings live in the section selected by `name` in `/HP21xx_IO_FPGA.cfg`; the loader and application share that file.

Before enabling HP service, confirm interface compatibility and establish backend readiness. Withdraw readiness when the backend is unavailable; DONE only says the FPGA configured, not that storage or the MCU application is ready. Respect card removal/write protect, DEV ownership and power-domain limits in normal operation and error recovery.

An FPGA-only profile still needs a safe MCU policy while its FPGA repurposes FSMC pins: leave the complete transport released and refrain from initializing the backend. Whether such profiles run a minimal management application or remain in the resident loader is open. They must never fall through to the installed disk application's normal FSMC initialization by accident.

Network-enabled applications need an Ethernet PHY/MAC driver, a suitable network stack and a persistent unique MAC-address policy. Place Ethernet DMA descriptors/buffers in DMA-accessible

SRAM, not CCM RAM; budget network buffers against SD/FSMC and application DMA use. Ethernet support need not be in the small resident SD loader. Experiment-specific DAC and spare-pin functions belong to the selected application; initialize them without violating released-transport or DEV ownership.

4.4 Other software

This section holds software material without a dedicated subsection in the requested outline, for the user's next allocation pass.

Recovery/UI and diagnostic policy

Define pendant/USB/LED messages for missing cards, locked cards, invalid profiles, incompatible image pairs, failed updates and DEV inhibition. Select timeouts, retry counts, diagnostic commands and a durable metadata format. Decide whether DEV also prevents automatic MCU application replacement; the required restriction concerns FPGA JTAG. Define the no-card and FPGA-only behavior and whether rollback/golden images are needed.

Verification and acceptance

Validate ROM USB DFU/USART recovery, SD switch semantics, descriptor/hash checking, unchanged-image boot without erase, interrupted application/FPGA updates, JTAG release and strict DEV behavior across resets/errors. Exercise USB-only startup without backpower/reset clamping and verify all 19 MCU FSMC pins remain released for expansion profiles. Hardware acceptance also needs rail/reset/flash-ready timing, receiver/driver loads, and HP bus timing/peripheral tests on the completed integrated board.

Remaining software decisions and verification work are listed in **section 5**.

5. Open issues

Only unresolved decisions and implementation/verification work are listed here. The component choices, pin map, SWD-only debug policy, HSWAPEN low, SUSPEND grounded and single configuration file above are the current design.

5.1 Hardware

ID	Section	Open issue
H01	3.4	Derive the final shaped outline/card-edge layout from the supplied KiCad board; close endpoint gaps, check fingers/plating, slot fit, mounting and cable/latch clearance. Verify maximum installed heights, including tolerances and shield tabs, within the 12.8 mm component-side envelope. Review any recessed jack's cutout and underside clearance separately.
H02	3.2.6 / 3.3.9	Size the wire-wrap area and choose posts/external connector contacts, using the fixed 1.6 mm pad / 0.8 mm drill geometry. Allocate fitted AUX1/AUX2 and spare RS-232 nodes in the connector pinout; their application functions remain undefined.
H03	3.2.1 / 3.3.1	Validate the selected buck LC parts, effective capacitance, layout, FPGA power estimate, startup/thermal behavior and per-slot power path. Establish HP output/pendant currents and USB current/inrush/suspend policy.

ID	Section	Open issue
H04	3.2.7 / 3.3.7	Validate reset gate/RC/button/diode operation and any additional rail supervision; verify PON/NRST/PROGRAM_B low levels, diode drops, flash readiness and USB-only startup without backpower or reset clamping.
H05	3.2.2 / 3.3.2	Qualify the shared 50 MHz oscillator/fan-out and proposed 8 MHz crystal/load capacitors; verify HSE/USB clock startup. Decide whether RTC crystal/battery support is needed.
H06	3.3.8	Verify RMII timing/jitter with the shared oscillator/fan-out; confirm MTJ-885X1 TME orderability and maximum height; validate H1102NLT external magnetics/centre-tap bias, proposed hardware POR, straps, MDIO bias, cable termination/ESD/isolation network, external LED wiring/current and 10/100 current measurements.
H07	3.2.3	Choose INIT_B/DONE/flash bias values and header footprint; verify startup pull-ups, bitstream pin settings and inactive HP states through reset, configuration and JTAG bridge operation.
H08	3.2.5	Validate receiver/driver thresholds, rise/fall times, transistor/ACT loading, resistor package dissipation and HP timing under real loads.
H09	3.3.3 / 3.3.4	Finalize USB/SD footprints, ESD/ground routing and any SD power switching; verify PB0 divider tolerances/leakage and SD switch behavior.
H10	3.3.5	Finalize the selected 4P4C jack placement, numbered pinout and cable; set baud rate and protected +5 V budget. Allocate spare RS-232 channel connector contacts and validate receiver level translation, power-domain isolation and all loaded drivers.
H11	3.4	Validate currents and visibility for the accepted LEDs and three regulator indicators; reconcile all provisional support quantities and exact parts against the completed schematic; finish component designators, assembly footprints and debugger/pendant mating clearances. Confirm JLCPCB orderability and all TME through-hole parts before purchase.

5.2 Software and validation

ID	Section	Open issue
S01	4.1	Implement the shared section-aware `HP21xx_IO_FPGA.cfg` parser and selected-section handoff; define bounds/defaults/error handling for the schema.
S02	4.2 / 4.2.1	Implement loader/application linking, descriptor placement and persistent metadata/valid markers; prove loader capacity, hash checks, unchanged-image boot and interrupted-update recovery.

ID	Section	Open issue
S03	4.2 / 4.4	Define the pendant selection protocol and controlled application/gateware switchover. Finalize recovery indication/timeouts, FPGA-only management/handoff behavior, and whether DEV also inhibits automatic MCU application replacement. Decide whether rollback/golden images are required.
S04	4.2.2	Implement GigaDevice flash support, a verified JTAG-to-SPI bridge, ID/hash checks, flash verification and reliable reboot. Define persistent-image verification after external JTAG programming.
S05	3.2.4 / 3.3.6 / 4.3	Define AUX1/AUX2 function/direction per application and enforce SWD-only startup, DEV prohibition and JTAG/FSMC/AUX ownership across handoff, errors and resets. Validate complete transport release for FPGA-only profiles.
S06	4.3	Define HP readiness/priority behavior during MCU reset, DEV, updates and backend failure; complete physical HP peripheral tests.
S07	3.3.3 / 4.4	Validate ROM DFU/USART/SWD recovery with pendant attached, independent MCU power, and no unintended FPGA activity.
S08	4.3	Implement Ethernet MAC/PHY/stack, unique MAC-address policy and DMA/SRAM budgets; leave experiment-specific analog/digital expansion functions to the applications that need them.

Appendix A. HP signal allocation

Directions below are relative to the new board: **IN = computer to board, OUT = board to computer**. The worksheet's IOB input/output names are from the computer perspective. Preserve separate physical receiver/driver channels. The table is transcribed from the two workbooks and the current controller pin-map reader; it is a compatibility baseline, not a manufactured connector footprint. HP connector numbering/orientation still needs mechanical review.

HP controls

Signal	HP connector pin	Direction	FPGA pad
PRL	3	OUT	P51
FLGL	4	OUT	P50
SFC	5	IN	P48
IRQL	6	OUT	P47
CLF	7	IN	P46
IEN	8	IN	P45
STF	9	IN	P44
IAK	10	IN	P43
T3	11	IN	P41

Signal	HP connector pin	Direction	FPGA pad
SKF	12	OUT	P40
CRS	13	IN	P35
SCM_L	14	IN	P34
IOG	15	IN	P33
SCL_L	16	IN	P32
POPIO	17	IN	P30
IOB16_OR_BIOS_N	18	IN	P29
SRQ	19	OUT	P27
IOO	20	IN	P26
CLC	21	IN	P24
STC	22	IN	P23
PRH	23	IN	P22
IOI	24	IN	P21
SFS	25	IN	P17
SIR	32	IN	P9
IRQH	33	OUT	P8
SCL_H	34	IN	P7
SCM_H	37	IN	P5
ENF	46	IN	P141
FLGH	49	OUT	P140
RUN	50	IN	P139
RUN_DRIVE	50	OUT	P138
EDT	62	IN	P123
PON	66	IN	P119
BIOO_N	67	IN	P118
SFSB_OR_BIOI_N	73	IN	P117

PON above is the inverted protocol receiver, separate from the conditioned configuration-reset path. RUN and RUN_DRIVE share HP pin 50 but use separate FPGA/receiver/driver channels; RUN drive remains inactive in the current image.

HP data buses

Bit	Computer -> board HP pin	FPGA receiver pad	Board -> computer HP pin	FPGA driver pad
0	26	P16	35	P6
1	29	P12	38	P2
2	30	P11	41	P1
3	64	P121	45	P142
4	77	P115	42	P57
5	80	P111	51	P137
6	81	P105	53	P133
7	84	P101	52	P134
8	27	P15	54	P132
9	28	P14	56	P55
10	31	P10	58	P127
11	60	P126	55	P131
12	78	P114	57	P58
13	79	P112	61	P124
14	82	P104	65	P120
15	83	P102	74	P116

HP power and unused contacts

Function	HP connector pins
Ground	1, 2, 85, 86
+5 V	39, 40
-2 V	47, 48
+12 V	43, 44
-12 V	69, 70
+28 V	36
Unused in the workbook	59, 63, 68, 71, 72, 75

References and evidence

- HP bus workbook and STM32 workbook.
- PON circuit photo; user-reported repeated successful starts, with no measured voltage/waveform margins supplied.

- FPGA implementation, pin-map generator, STM32 transport, and disk backend.
- Prototype operational history. New-board intent in this document does not change installed firmware or prototype wiring.
- External primary sources are linked at the relevant requirements above.